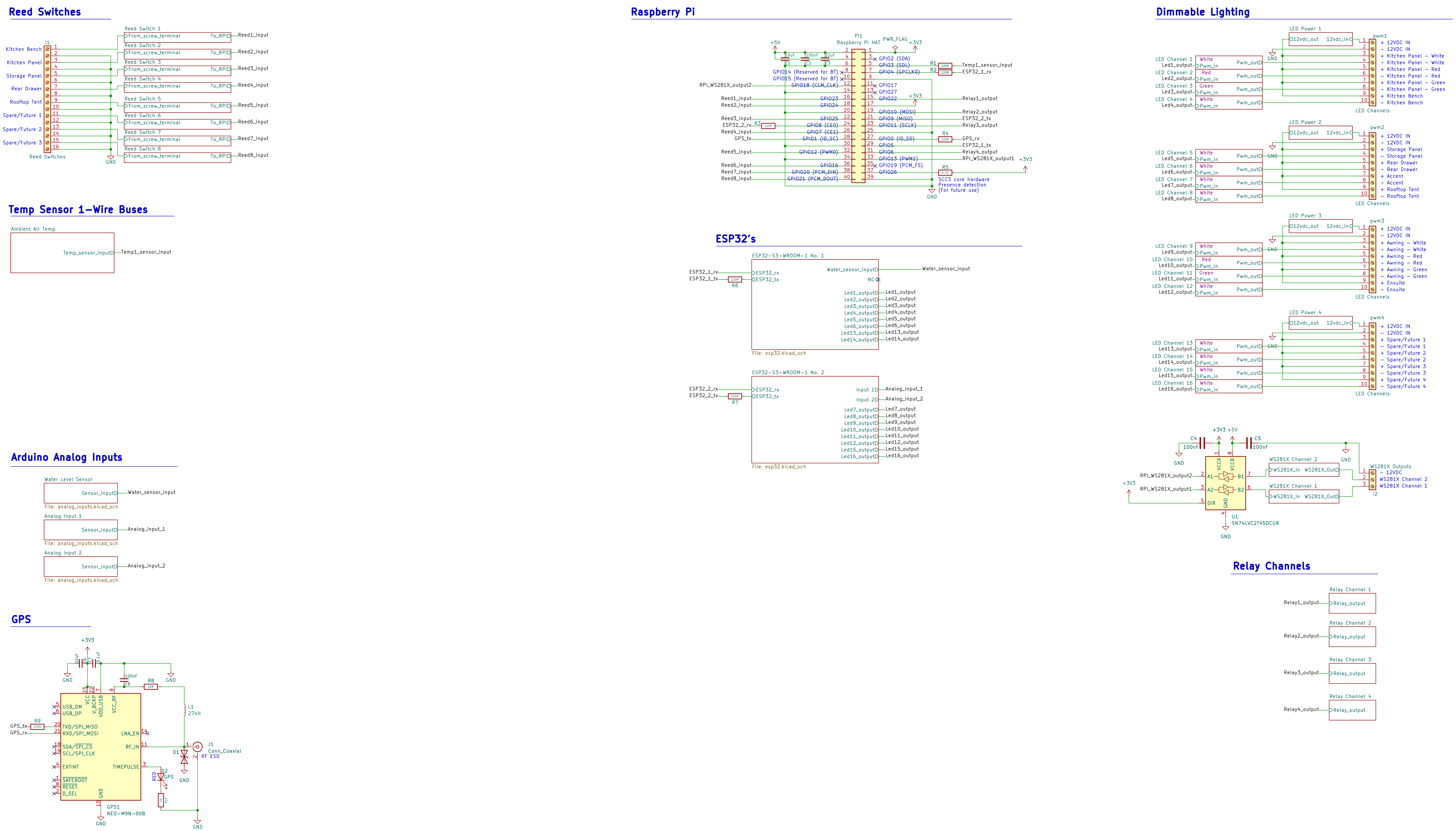
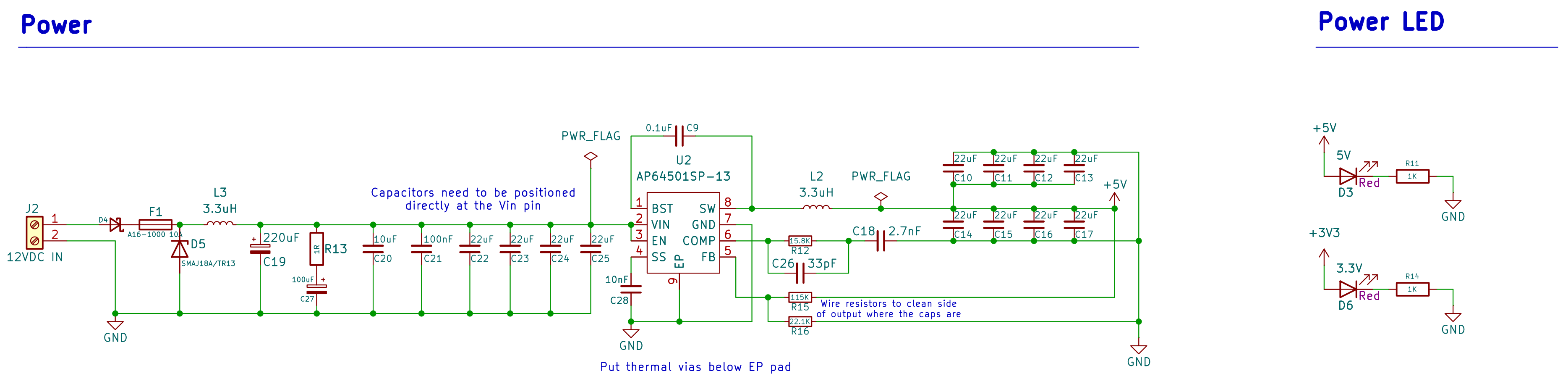


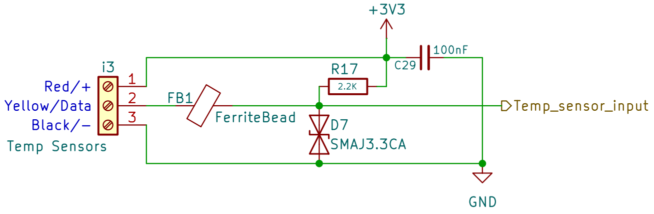


Notes

1. Power circuit provides 5V
2. RPI Provides 3.3V
3. 1-Wire Bus Notes:
   Board has a 2.2K pull-up on the data line.
   Put a 220R resistor on the data pin of each temp sensor but ensure it's not in series with the bus.
   Put a 100nF ceramic cap across VCC and GND at each temp sensor.
4. WS281X RPI Notes:
   Put a 220uF electrolytic cap between 12V and GND at the start of each LED strip
5. UARTS:
   UART1 / GPIO14/15 / Reserved for Bluetooth
   UART2 / GPIO00/01 / GPS
   UART3 / GPIO04/05 / ESP32 No. 1
   UART4 / GPIO08/09 / ESP32 No. 2
6. 12V LED Strip Notes (not required but good practice):
   Put a 220uF 25V electrolytic cap across 12V and GND at each LED strip.
   Put a 100nF 25V ceramic cap across 12V and GND at each LED strip.

Scale: NTS
Drawn By: Joel Goudswaard
Singularity Automation Pty Ltd
Sheet: /
File: SCCS-RPI.kicad_sch
Title: Singularity Camper Control System (SCCS)

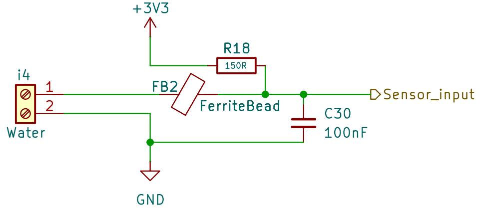
Size: A2	Date: 2026-08-10	**Rev: 1.1**
KiCad E.D.A. 10.0.4		Id: 1/41



Sheet: /Ambient Air Temp/
File: 1_wire_bus.kicad_sch

Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 2/41

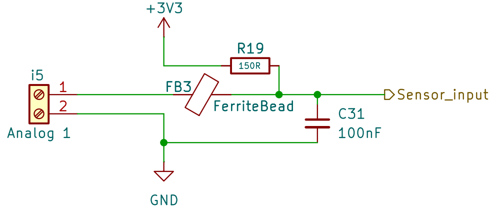


Sheet: /Water Level Sensor/
File: analog_inputs.kicad_sch

Title:

Size: A5 | Date: | **Rev:**

KiCad E.D.A. 10.0.4 | Id: 3/41



Sheet: /Analog Input 1/
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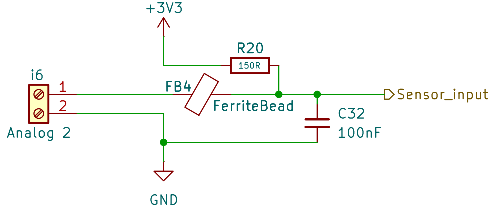
Title:

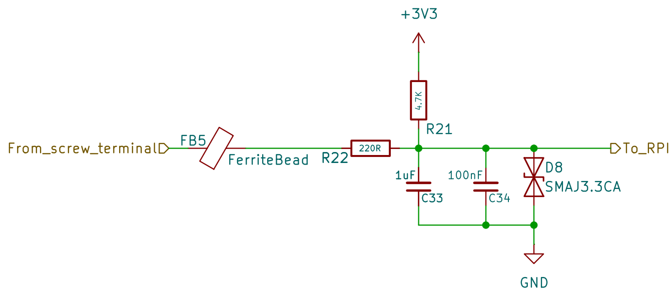
Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 4/41



Sheet: /Analog Input 2/
File: analog_inputs.kicad_sch

Title:

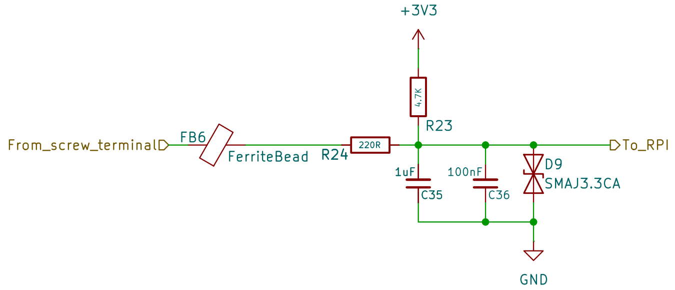
Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 5/41



Sheet: /Reed Switch 1/
File: reed_input.kicad_sch

Title:

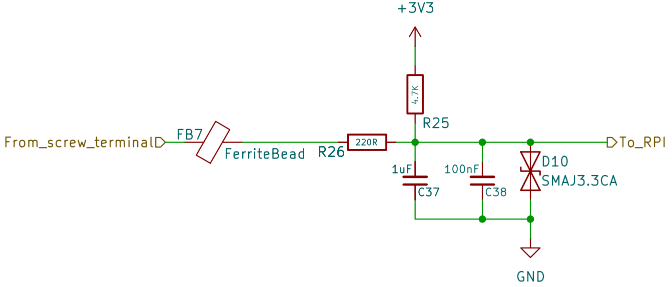
Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 6/41



Sheet: /Reed Switch 2/
File: reed_input.kicad_sch

Title:

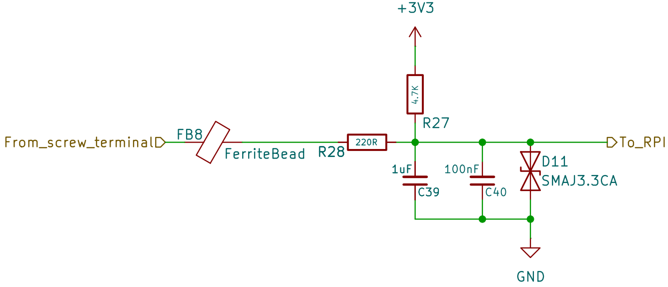
Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 7/41



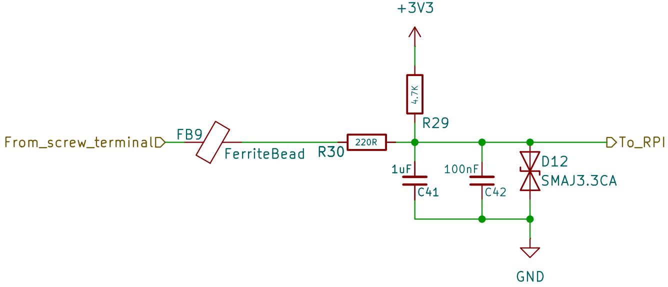
Sheet: /Reed Switch 3/
File: reed_input.kicad_sch

Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 8/41



Sheet: /Reed Switch 4/		
File: reed_input.kicad_sch		
Title:		
Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 9/41



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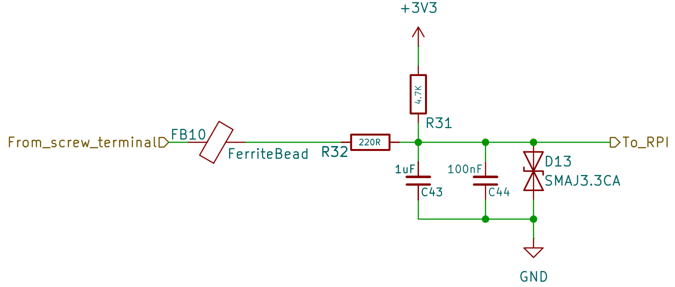
Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 10/41



Sheet: /Reed Switch 6/
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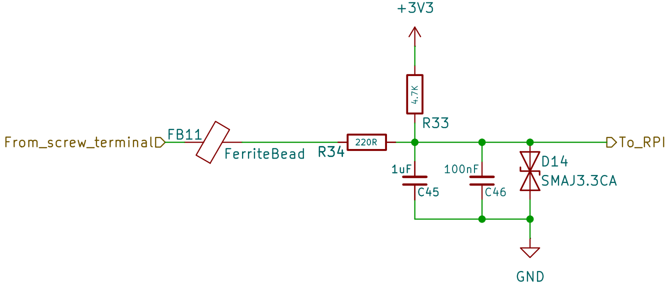
Title:

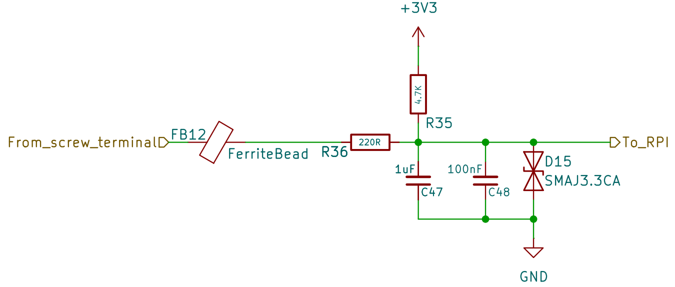
Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 11/41



Sheet: /Reed Switch 7/
File: reed_input.kicad_sch

Title:

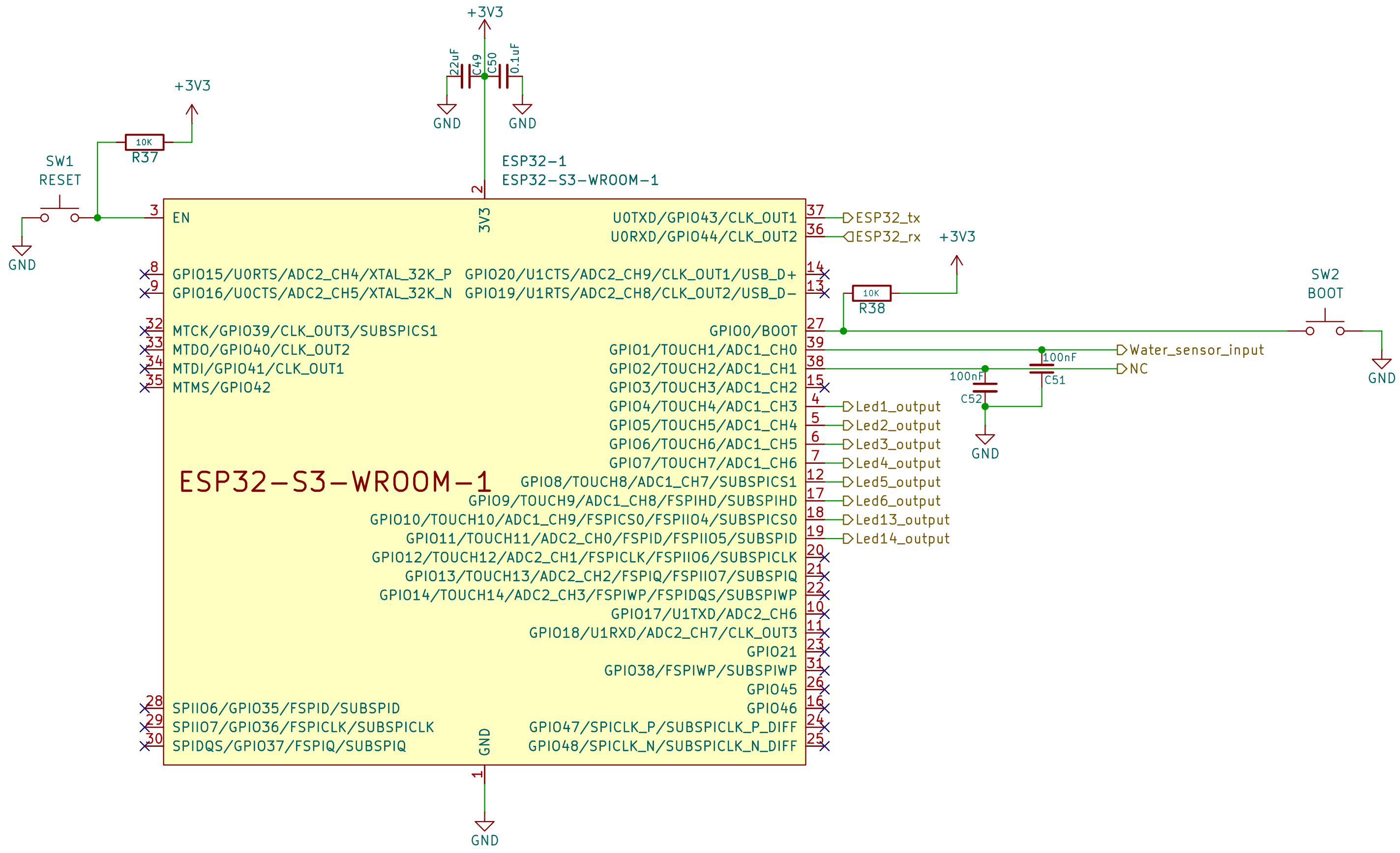
Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 12/41



Sheet: /Reed Switch 8/
File: reed_input.kicad_sch

Title:

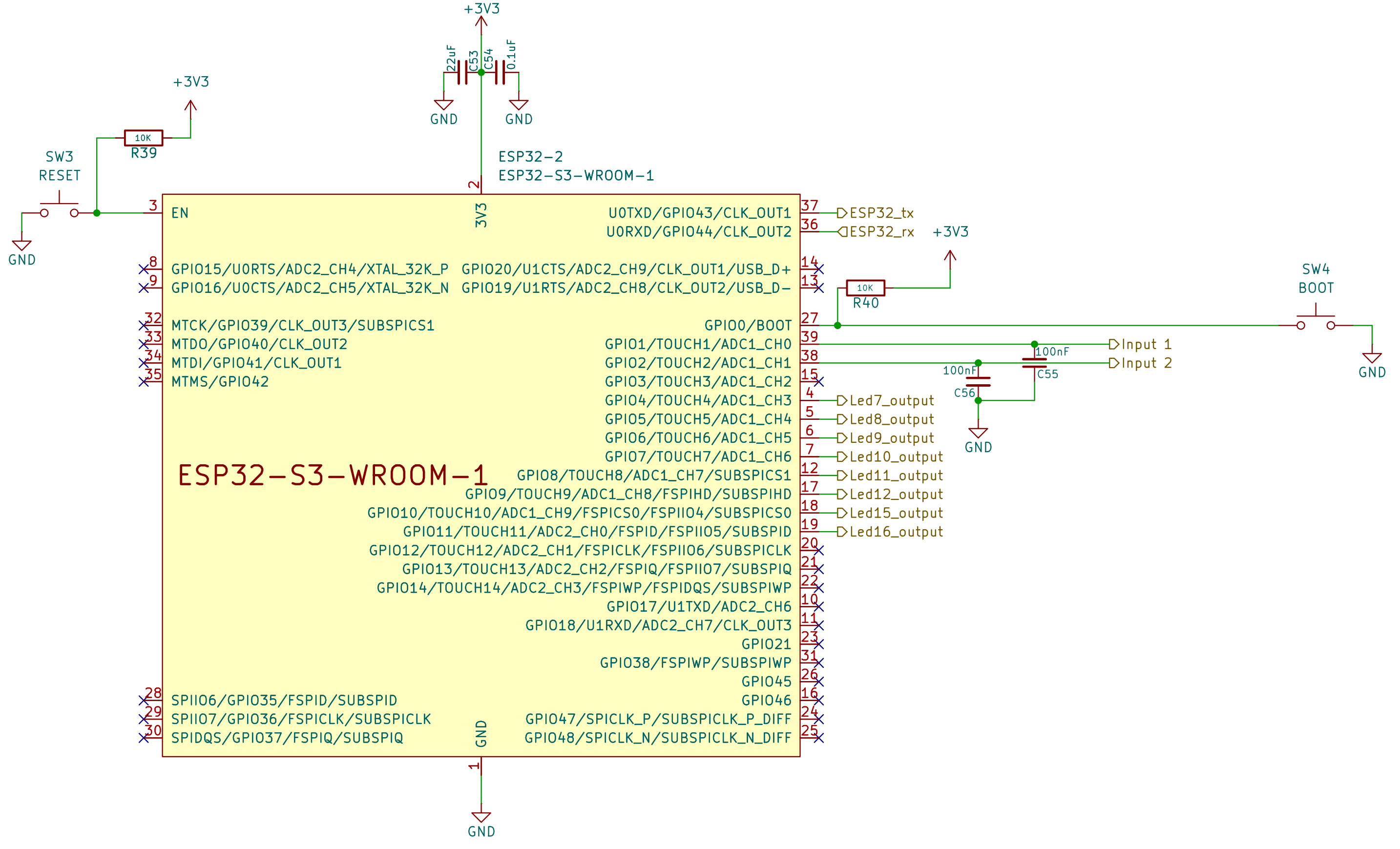
Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 13/41



Sheet: /ESP32-S3-WROOM-1 No. 1/
File: esp32.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 14/41

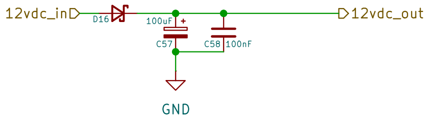


Sheet: /ESP32-S3-WROOM-1 No. 2/
File: esp32.kicad_sch

Title:

Size: A4 | Date: | **Rev:**

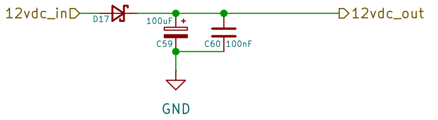
KiCad E.D.A. 10.0.4 | Id: 15/41



Sheet: /LED Power 1/
File: led_power.kicad_sch

Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 16/41



Sheet: /LED Power 2/
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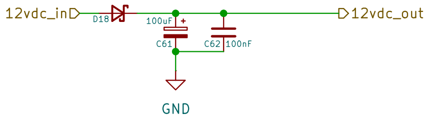
Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 17/41



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Title:

Size: A5	Date:	**Rev:**
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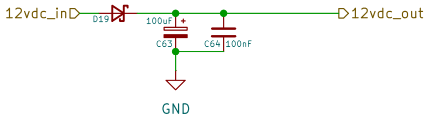

Sheet: /LED Power 4/
File: led_power.kicad_sch

Title:

Size: A5	Date:	**Rev:**
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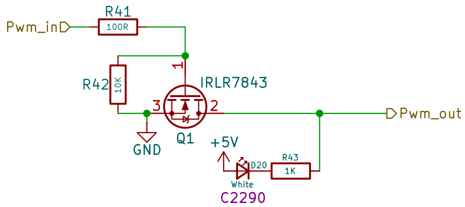
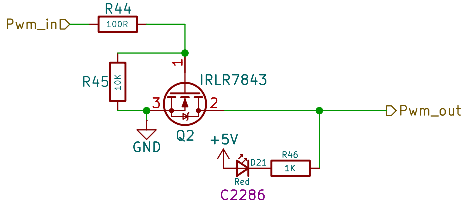


Sheet: /LED Channel 1/
File: led_channel.kicad_sch

Title:

Size: A5 Date: **Rev:**

KiCad E.D.A. 10.0.4 Id: 20/41



Sheet: /LED Channel 2/
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Title:

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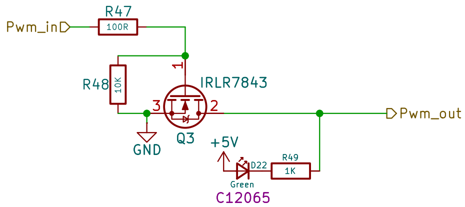
KiCad E.D.A. 10.0.4		Id: 21/41



Sheet: /LED Channel 3/
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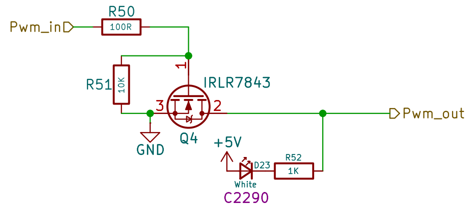
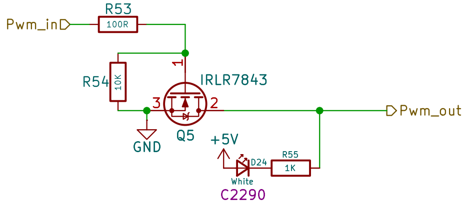


Sheet: /LED Channel 4/
File: led_channel.kicad_sch

Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 24/41



Sheet: /LED Channel 5/
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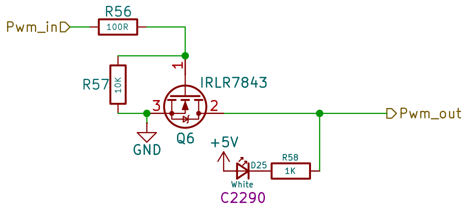
Title:

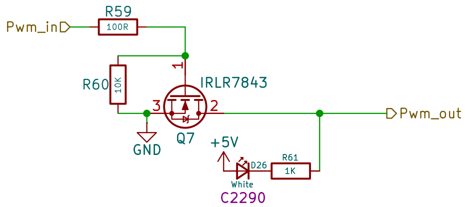
Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 25/41



Sheet: /LED Channel 6/
File: led_channel.kicad_sch

Title:

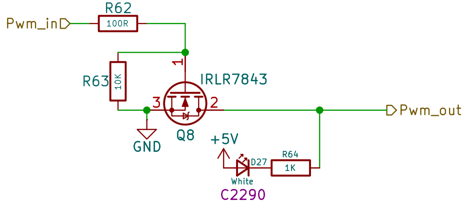
Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 26/41



Sheet: /LED Channel 7/
File: led_channel.kicad_sch

Title:

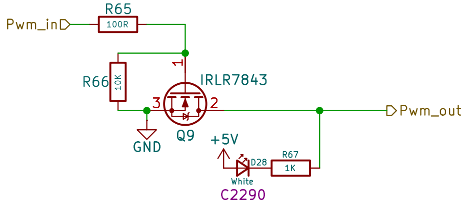
Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 27/41



Sheet: /LED Channel 8/
File: led_channel.kicad_sch

Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 28/41



Sheet: /LED Channel 9/
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Title:

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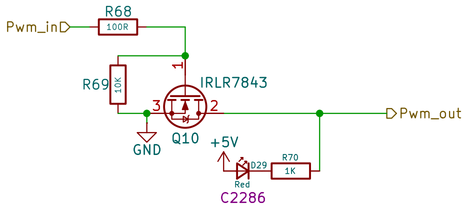

Sheet: /LED Channel 10/
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Title:

Size: A5	Date:	**Rev:**
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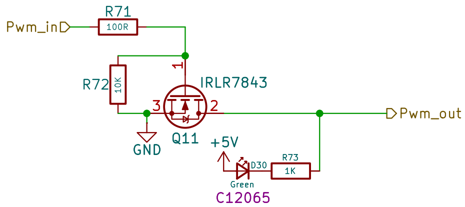

Sheet: /LED Channel 11/
File: led_channel.kicad_sch

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Size: A5	Date:	**Rev:**
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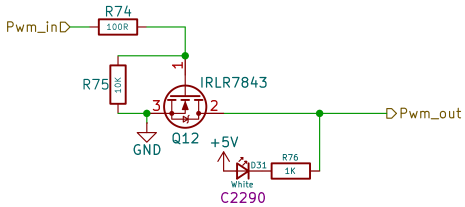

Sheet: /LED Channel 12/
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Size: A5	Date:	**Rev:**
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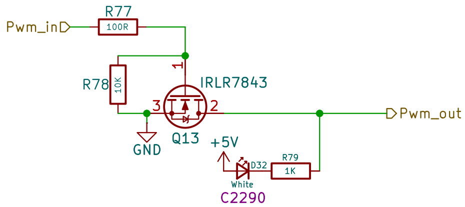

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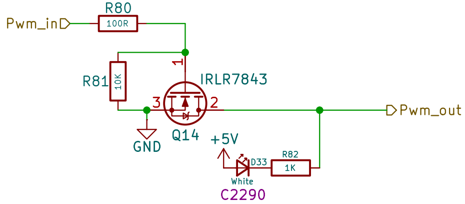

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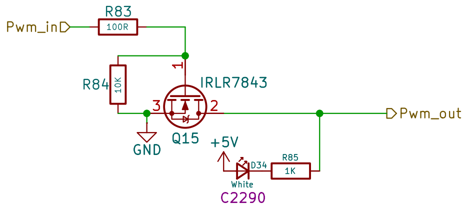

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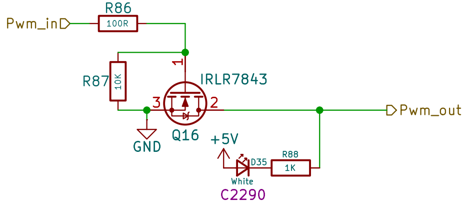

Sheet: /LED Channel 16/
File: led_channel.kicad_sch

Title:

Size: A5	Date:	**Rev:**
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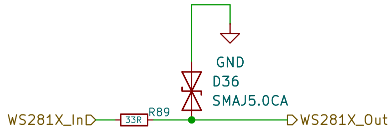
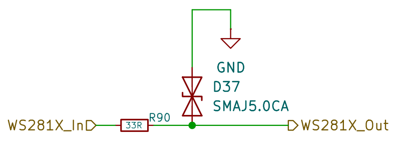


Sheet: /WS281X Channel 1/
File: ws2815.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 37/41



Sheet: /WS281X Channel 2/
File: ws2815.kicad_sch

Title:

Size: A4	Date:	**Rev:**
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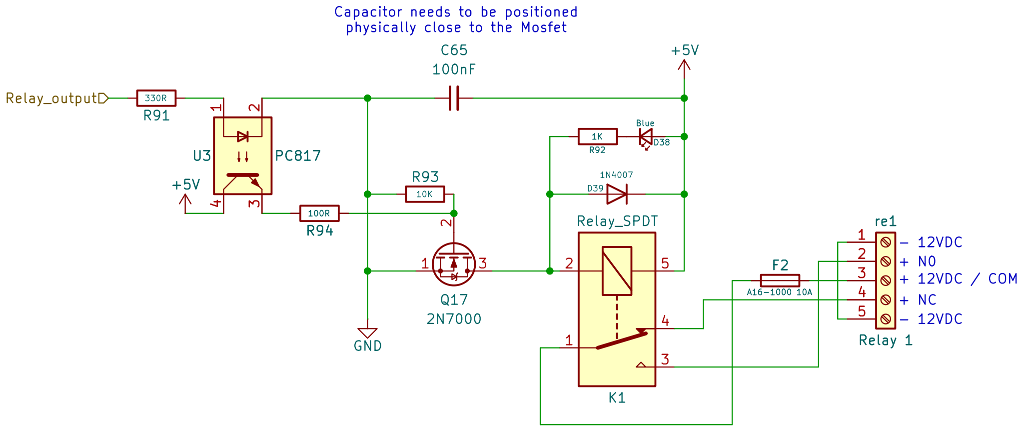

Sheet: /Relay Channel 1/
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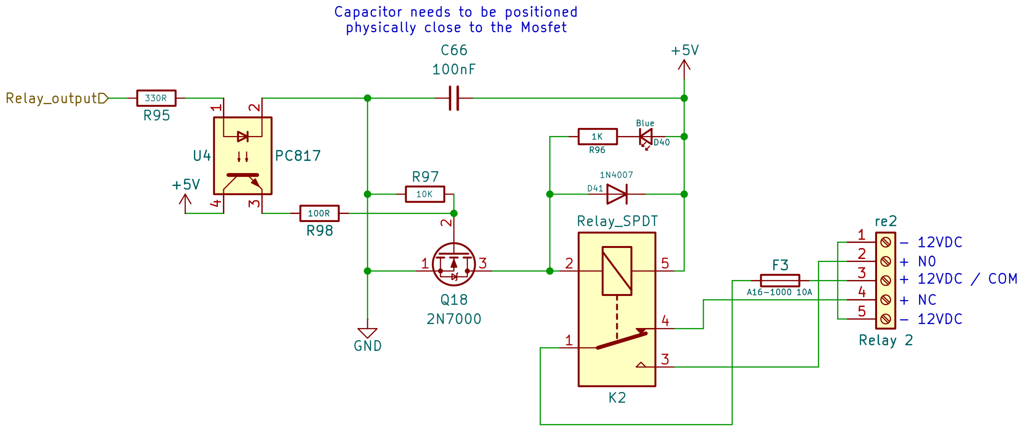
Title:

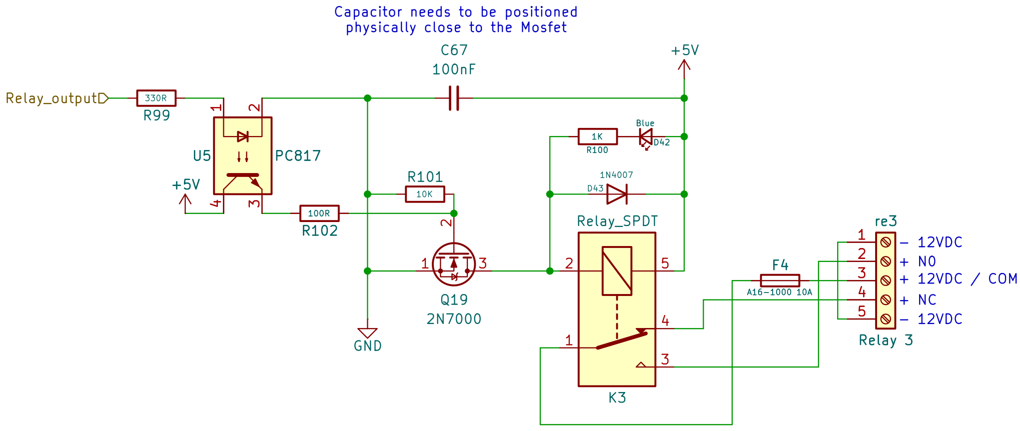
Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 39/41



Sheet: /Relay Channel 2/
File: relay_channel.kicad_sch

Title:

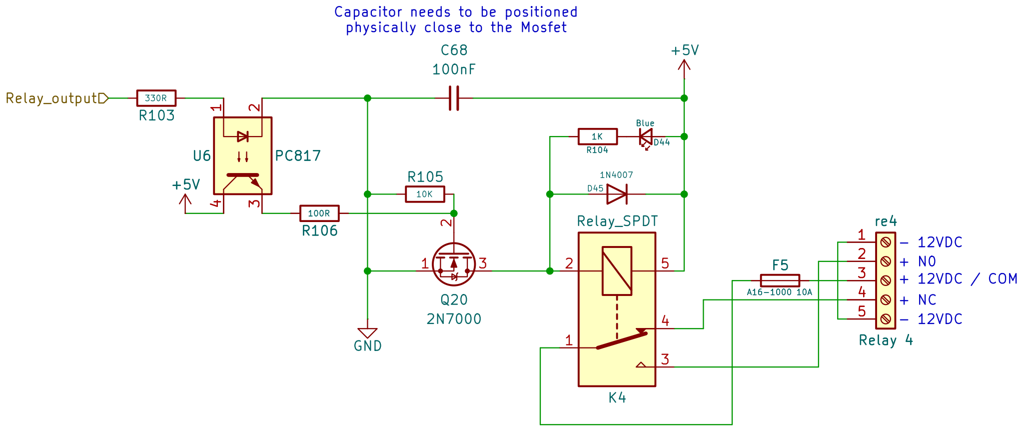
Size: A5	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 40/41



Sheet: /Relay Channel 3/
File: relay_channel.kicad_sch

Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 41/41



Sheet: /Relay Channel 4/
File: relay_channel.kicad_sch

Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 42/41